Configuration and secrets

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

Make the distinction between a setting and a credential concrete.

Teach required-key diagnostics and allowlisted output using mock values and narrowly checked claims.

Keep one habit: Model startup reasoning → inspect emitted records → diagnose output-channel leaks → assess fresh browser-error reasoning → revisit a missing setting.

MAIN LESSON · SECTION 1

Prepare the connected service investigation

> Keep modelling, shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Explain the **mechanism** and terminology of **configuration** and **secret**s

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Opening retrieval

Without opening the old lesson, explain one key idea from Typed contracts and schemas. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of typed contracts and schemas

Reference example, not the only acceptable wording: Teacher explanation: Typed contracts define values, required fields and failure outcomes. Runtime **input** still needs validation even when program annotations exist.

Accept exactly room_id as a nonempty string and requested_count as a nonnegative integer excluding bool. Reject extra fields, missing fields and wrong types before **process**ing.

Without opening the old lesson, explain one key idea from Requirements and service boundaries. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of requirements and service boundaries

Reference example, not the only acceptable wording: Teacher explanation: A service boundary states who can request what, what data may be read and which actions are outside scope. Requirements must be observable.

The fictional student timetable service reads a public schedule and returns a meeting time or review status. It cannot edit schedules or email users. Test success and refusal before adding model behaviour.

Pictured end goal

Explain **configuration** and **secret**s using the deciding step in this fictional service.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Typed contracts and schemas

Requirements and service boundaries

Use the three-session teaching cycle above, with extra time as needed. **Configuration** supplies non-**secret** settings; **secret**s provide privileged values. Keep **secret**s out of source, logs and client responses. Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions. Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the **mechanism** and terminology of **configuration** and **secret**s

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach classification, startup validation and output selection together

> Use only invented placeholders and inspect exactly the records the mock emits.

Definitions and complete input/output contract

Configuration supplies non-**secret** settings; **secret**s provide privileged values. Keep **secret**s out of source, logs and client responses.

The toy service reads DATA_VERSION from **configuration** and a signing **secret** from server-only **input**. Validate required settings at startup and fail clearly if absent; do not substitute a published default **secret**.

Input or record	Declared role and rule
DATA_VERSION	Reviewed non-secret configuration: nonblank string
MAX_WORDS	Reviewed non-secret configuration: exact positive integer, bool excluded
SIGNING_SECRET	Server-only mock input: nonblank placeholder; never copy its value into emitted records
Missing/invalid required field	Block startup and report the key name, not its value
Successful public record	Only status, data_version and maximum_words
Internal runtime state	May carry the placeholder separately; never print or serialize that state in the lab
Actual operations	No real environment, credential store, signing, model answer or deployment is used

Startup flow and main debug-output reasoning

Check required values, then build a separate public record

STAGE	LOCAL CLASSROOM BEHAVIOUR
1. Read declared inputs	Take DATA_VERSION/MAX_WORDS from a mock config dict; placeholder from a separate server-only dict
2. Validate required values	Missing or invalid keys → startup_blocked with names only; no default credential
3. Hold internal state	Keep the placeholder on the internal path, separate from displayed status
4. Build an allowlisted public record	Ready status + reviewed data version + configured word limit; never dump all inputs
5. Inspect emitted records	Check that the tested mock private values do not appear in any serialized public/error record

This is a local handling exercise. It demonstrates chosen output fields, not a production secret-management system.

Ask learners to name the missing key and the minimum diagnostic needed. Do not let them dump real environment variables or bring real credentials into this exercise. The provided code reads only mock dictionaries and checks chosen emitted records. Public configuration

must actually have been classified as non-secret; an allowlisted field name alone cannot make a sensitive value public.

A configured word limit is not proof that responses obey it. Keep mock startup success, tested disclosure checks and absent production evidence separate.

Configuration and secrets — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
A startup script prints every environment variable while debugging a missing data version.	The toy service reads DATA_VERSION from configuration and a signing secret from server-only input. Validate required settings at startup and fail clearly if absent; do not substitute a published default secret.	It risks exposing secrets unrelated to the error. Report the missing key name without values and validate only required fields.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: It risks exposing **secret**s unrelated to the error. Report the missing key name without values and validate only required fields.

Qualification: A classroom mock is not a production credential-management design; no real **secret**s are needed here.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

A startup script prints every environment variable while debugging a missing data version.

It risks exposing **secret**s unrelated to the error. Report the missing key name without values and validate only required fields.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Complete missing, type and private-output probes

Complete mock probe	Expected record or scope
DATA_VERSION classroom-v1; MAX_WORDS 80; placeholder present	ready; public version and limit only
DATA_VERSION missing	startup_blocked; missing_or_invalid [DATA_VERSION], without input values
Server-only placeholder absent	startup_blocked; missing_or_invalid [SIGNING_SECRET], without a default
MAX_WORDS True, zero or a quoted number	startup_blocked under the exact positive-integer rule
Limit changes 80 → 60	Public configured limit changes; placeholder unchanged
Unrelated mock private value exists	Never iterate/dump it into the public/error record
Private store exists	Does not by itself prevent leaks in other logs, screenshots or code paths
Real credentials or deployment	Outside this mock; no real secrets or external operations needed

Separate picture inputs and configured-limit scope

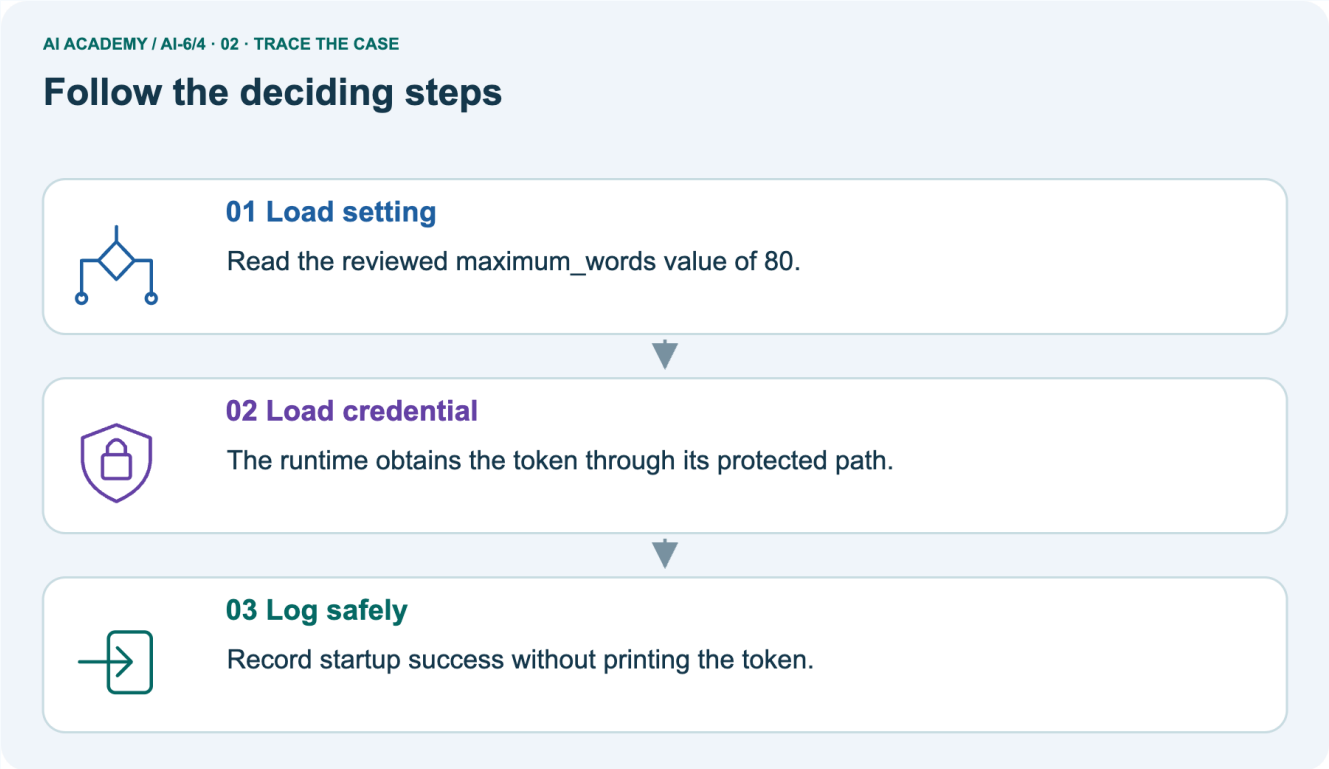
Settings and a sealed credential. A classroom bot needs a public answer-length setting and a private access token.

maximum_words=80 is public **configuration**. A separate protected store holds a placeholder credential.

The main failure is a debug script that prints every environment value while diagnosing a missing DATA_VERSION. Replace that output with a missing-key-name record. The picture separately sets maximum_words to 80 with a placeholder credential on a protected path. Changing the configured limit to 60 changes the reviewed setting; it does not require changing the credential.

Our local code records that setting only: no answer is generated, so enforcement of an actual response length has not been measured.

Picture trace and unchanged-placeholder key



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Load setting	Read the reviewed maximum_words value of 80.
2. Load credential	The runtime obtains the token through its protected path.
3. Log safely	Record startup success without printing the token.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Configuration controls behaviour. A credential grants access and needs different storage and handling.

Protected storage alone cannot prevent leaks through logs, screenshots or excessive runtime permissions.

Change: Change maximum_words to 60.

Prediction: The configured answer limit becomes 60; the private token stays unchanged.

Reason: An ordinary length setting does not require replacing the credential.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Secret	A value whose disclosure can grant privileged access
Configuration	Declared settings controlling operation
Redaction	Removing sensitive values from displayed or recorded output

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

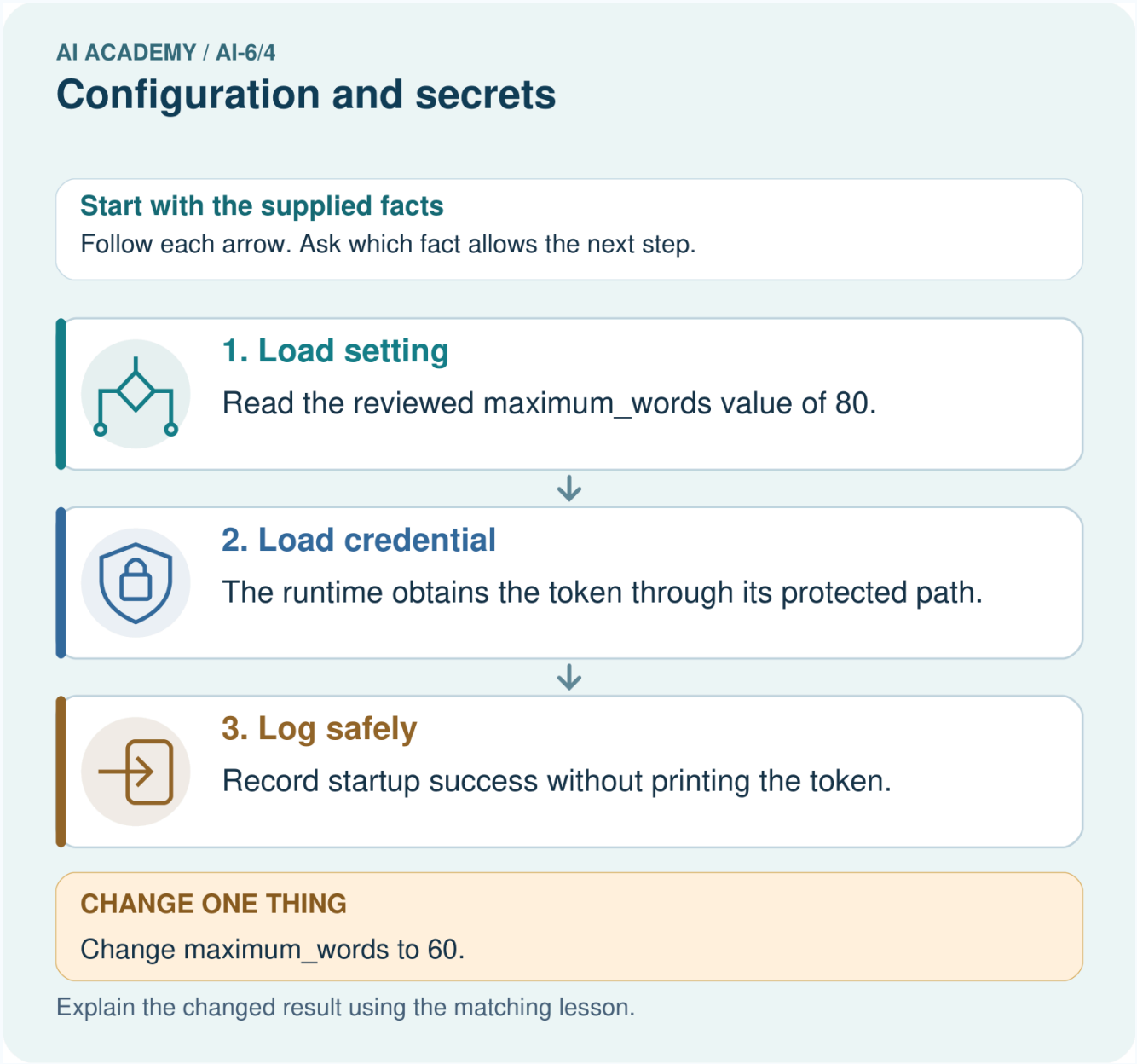
These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Configuration and secrets

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: A startup script prints every environment variable while debugging a missing data version. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: It risks exposing **secret**s unrelated to the error. Report the missing key name without values and validate only required fields.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for **Configuration** and **secret**s. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: A classroom mock is not a production credential-management design; no real **secret**s are needed here.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Point to the public setting and the protected credential. What changes when maximum_words becomes 60, and which information must never appear in the startup log?

Read the reviewed maximum_words value of 80. The runtime obtains the token through its protected path. Record startup success without printing the token.

Changed case: The configured answer limit becomes 60; the private token stays unchanged. An ordinary length setting does not require replacing the credential.

Changed-case reasoning: The configured answer limit becomes 60; the private token stays unchanged. An ordinary length setting does not require replacing the credential.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main missing-version diagnosis needs key-name-only output, not an environment dump. Mock valid startup ready; missing version/placeholder blocks without defaults; invalid MAX_WORDS blocks. Only reviewed non-secret fields go to public output.

Independent browser error: server-only credential crossed into a client response; replace it with a non-sensitive error identifier/status and keep any diagnostic redacted, without repeating the value. Preserve a record of the channel failure, not a copy of the credential. No real credential exercise or production-management guarantee.

Diagnose the current misconceptions

The worked result establishes every future case: A classroom mock is not a production credential-management design; no real **secret**s are needed here.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows	Next teaching action
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new service scenario

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Server **secret** exists, but a browser error includes its full value. What failed and what record should be produced instead?

Teacher explanation: The server/client boundary leaked the **secret**. Remove value disclosure; record a non-sensitive error ID and missing/invalid key name, and follow the actual **secret**-response procedure.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: **Configuration** supplies non-**secret** settings; **secret**s provide privileged values. Keep **secret**s out of source, logs and client responses.

The toy service reads DATA_VERSION from **configuration** and a signing **secret** from server-only **input**. Validate required settings at startup and fail clearly if absent; do not substitute a published default **secret**.

W2 Trace the worked case

A startup script prints every environment variable while debugging a missing data version. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: It risks exposing **secret**s unrelated to the error. Report the missing key name without values and validate only required fields.

W3 Audit the limit

A report omits this qualification: A classroom mock is not a production credential-management design; no real **secret**s are needed here.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: A classroom mock is not a production credential-management design; no real **secret**s are needed here.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. A classroom mock is not a production credential-management design; no real **secret**s are needed here.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: It risks exposing **secret**s unrelated to the error. Report the missing key name without values and validate only required fields.

Qualification: A classroom mock is not a production credential-management design; no real **secret**s are needed here.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

Server **secret** exists, but a browser error includes its full value. What failed and what record should be produced instead?

Teacher explanation: The server/client boundary leaked the **secret**. Remove value disclosure; record a non-sensitive error ID and missing/invalid key name, and follow the actual **secret**-response procedure.

W5 Design a revealing follow-up

Independent assessment and fresh retry

W4: The server/client boundary leaked the **secret**. Remove value disclosure; record a non-sensitive error ID and missing/invalid key name, and follow the actual **secret**-response procedure.

Require intermediate reasoning and the stated scope. Fresh retry: A debug log stores an access token to explain invalid **configuration**.

Expected reasoning: Redact the token; log the validation category and trace ID rather than credential contents.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and revisit service design

> Use saved evidence and boundary decisions to choose the next step.

Project and next-week bridge

Add a labelled **configuration** and **secret**s **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of **configuration** and **secret**s.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Authentication and authorisation. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Delayed return and support record

Suggested return points in this level: ai-6/5, ai-6/7, ai-6/10. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Explain the **mechanism** and terminology of **configuration** and **secret**s

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: A startup script prints every environment variable while debugging a missing data version. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: It risks exposing **secret**s unrelated to the error. Report the missing key name without values and validate only required fields.

Check every learner before discussion: Use the worked case for **Configuration** and **secret**s. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: A classroom mock is not a production credential-management design; no real **secret**s are needed here.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Change maximum_words to 60. Require a **prediction** and a reason before comparing. Expected result: The configured answer limit becomes 60; the private token stays unchanged.

Reason: An ordinary length setting does not require replacing the credential.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Configuration and secrets: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

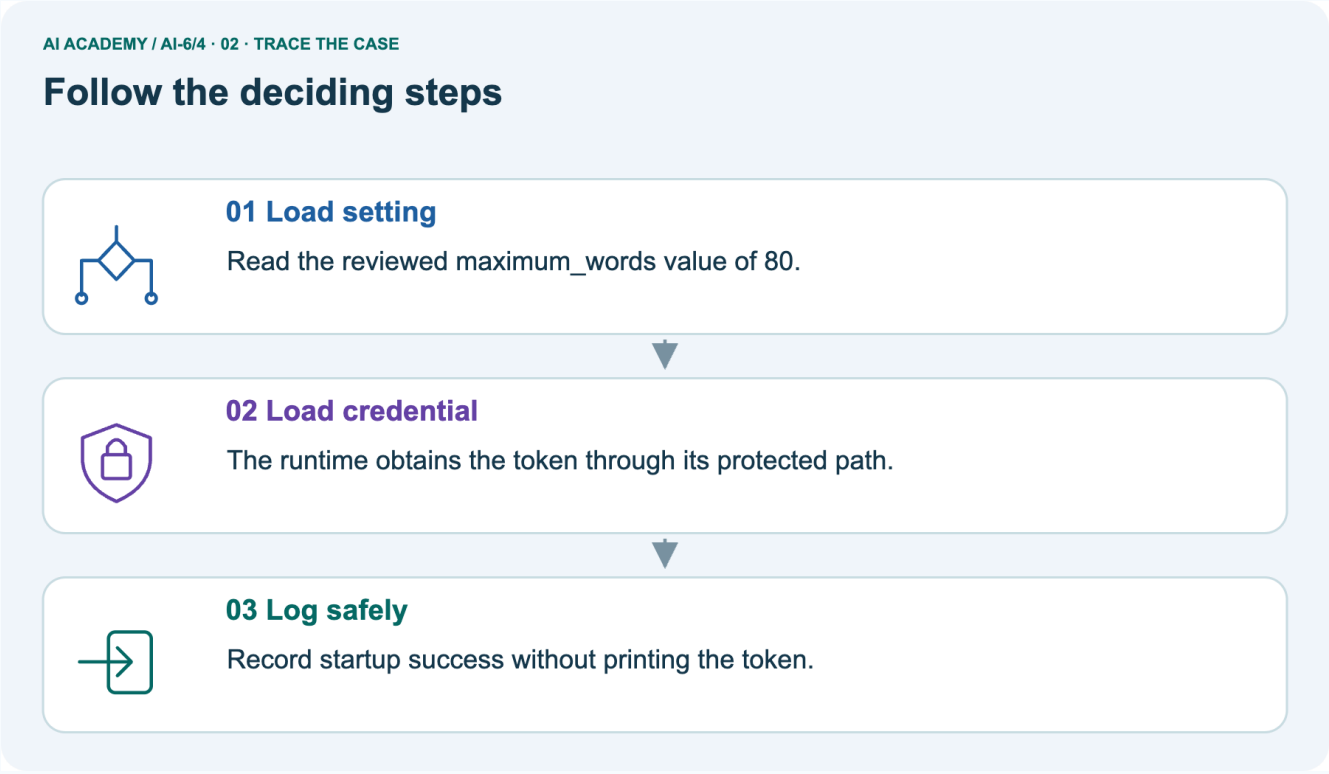
Settings and a sealed credential. A classroom bot needs a public answer-length setting and a private access token. Supplied facts: maximum_words=80 is public **configuration**.

A separate protected store holds a placeholder credential.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



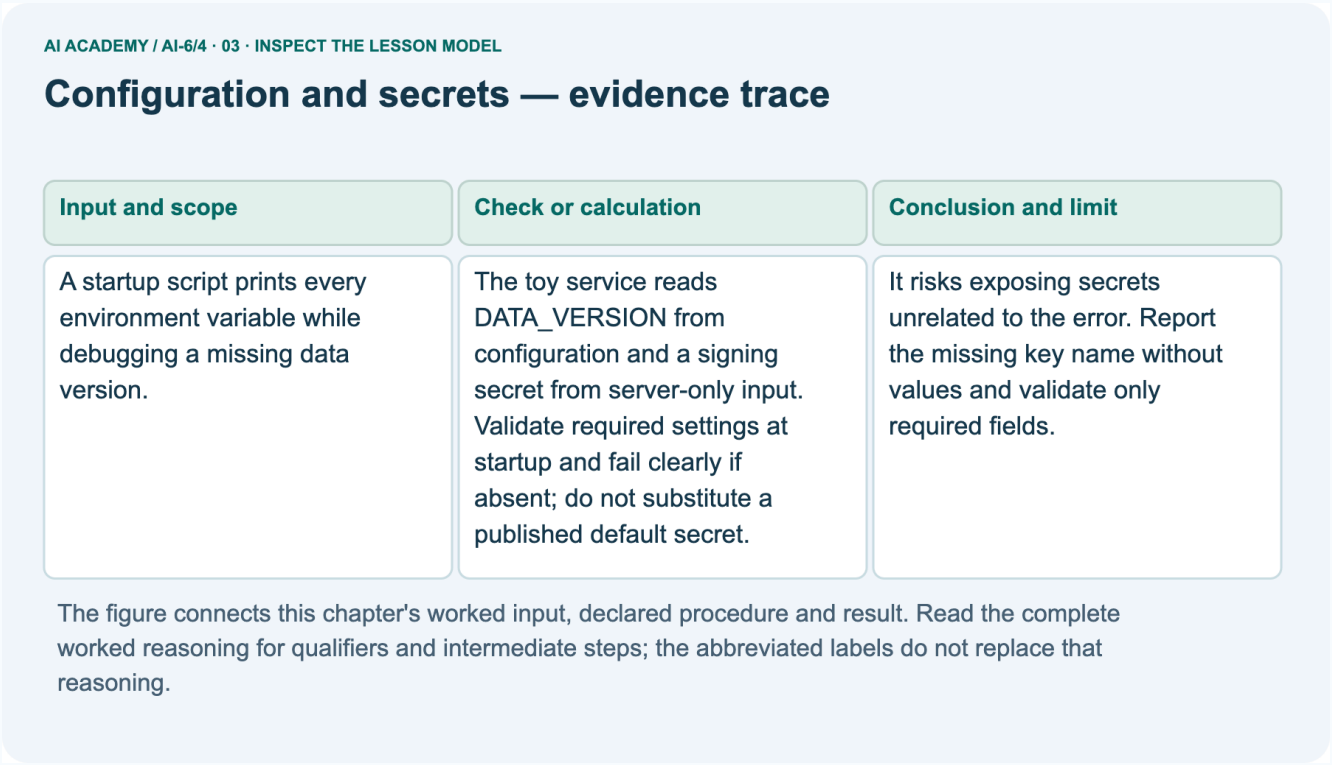
Follow the labelled picture with the matching explanation. Open the vector diagram

Load setting: Read the reviewed maximum_words value of 80. → Load credential: The runtime obtains the token through its protected path. → Log safely: Record startup success without printing the token.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Configuration and secrets — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Configuration and **secret**s — **evidence** trace.

A startup script prints every environment variable while debugging a missing data version. | The toy service reads DATA_VERSION from **configuration** and a signing **secret** from server-only **input**.

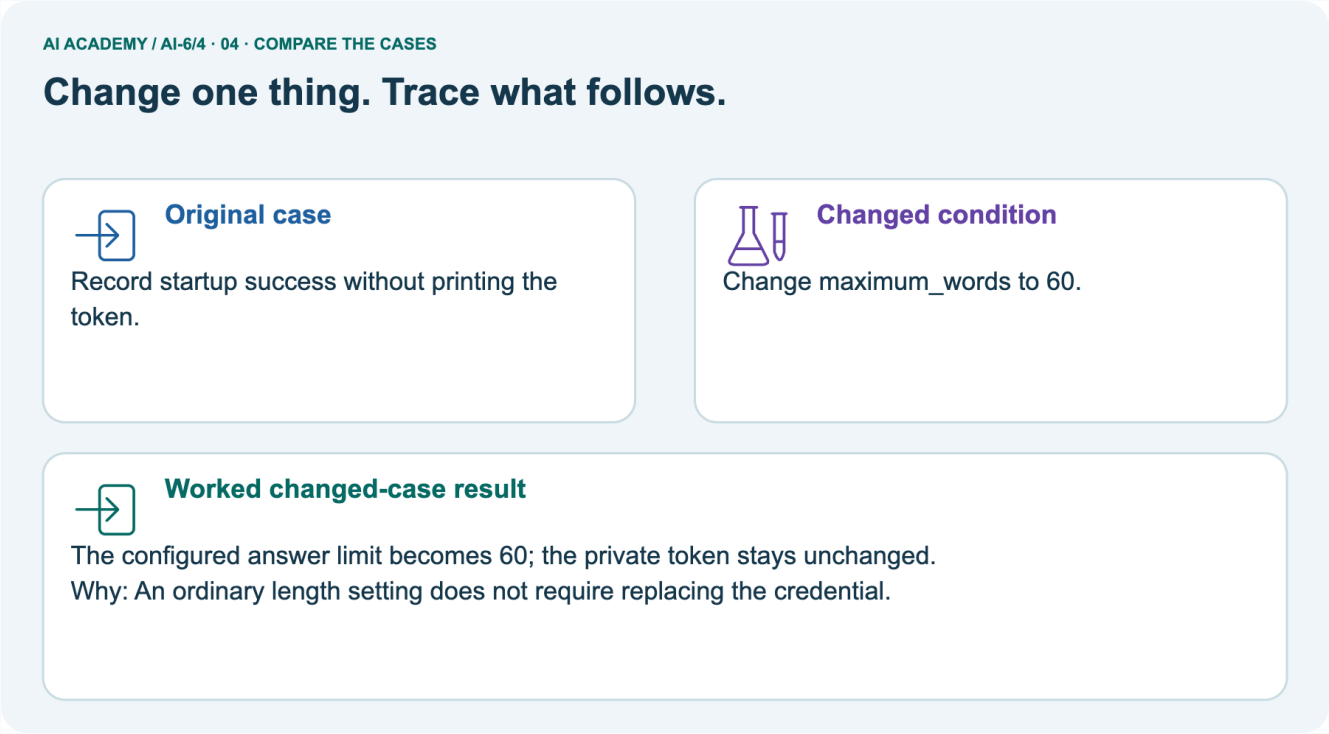
Validate required settings at startup and fail clearly if absent; do not substitute a published default **secret**. | It risks exposing **secret**s unrelated to the error.

Report the missing key name without values and validate only required fields.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Record startup success without printing the token. Changed condition: Change maximum_words to 60. Expected result: The configured answer limit becomes 60; the private token stays unchanged.

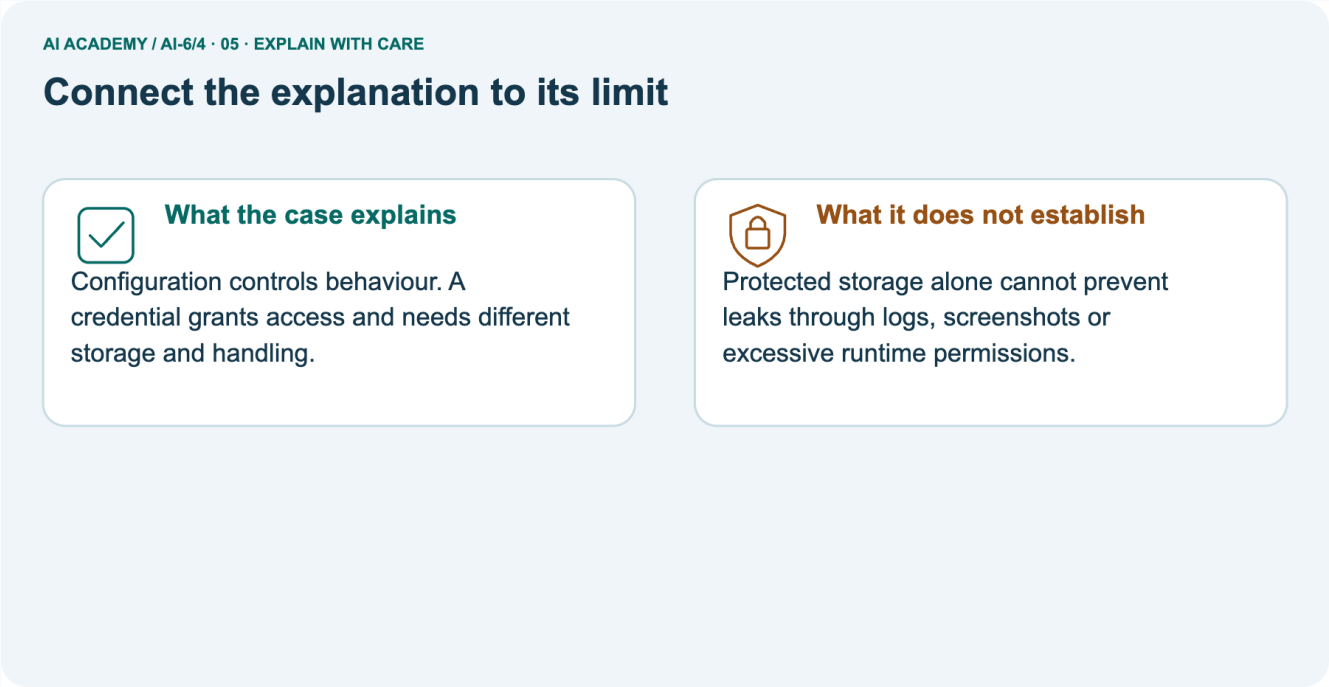
Reason: An ordinary length setting does not require replacing the credential.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: An ordinary length setting does not require replacing the credential.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

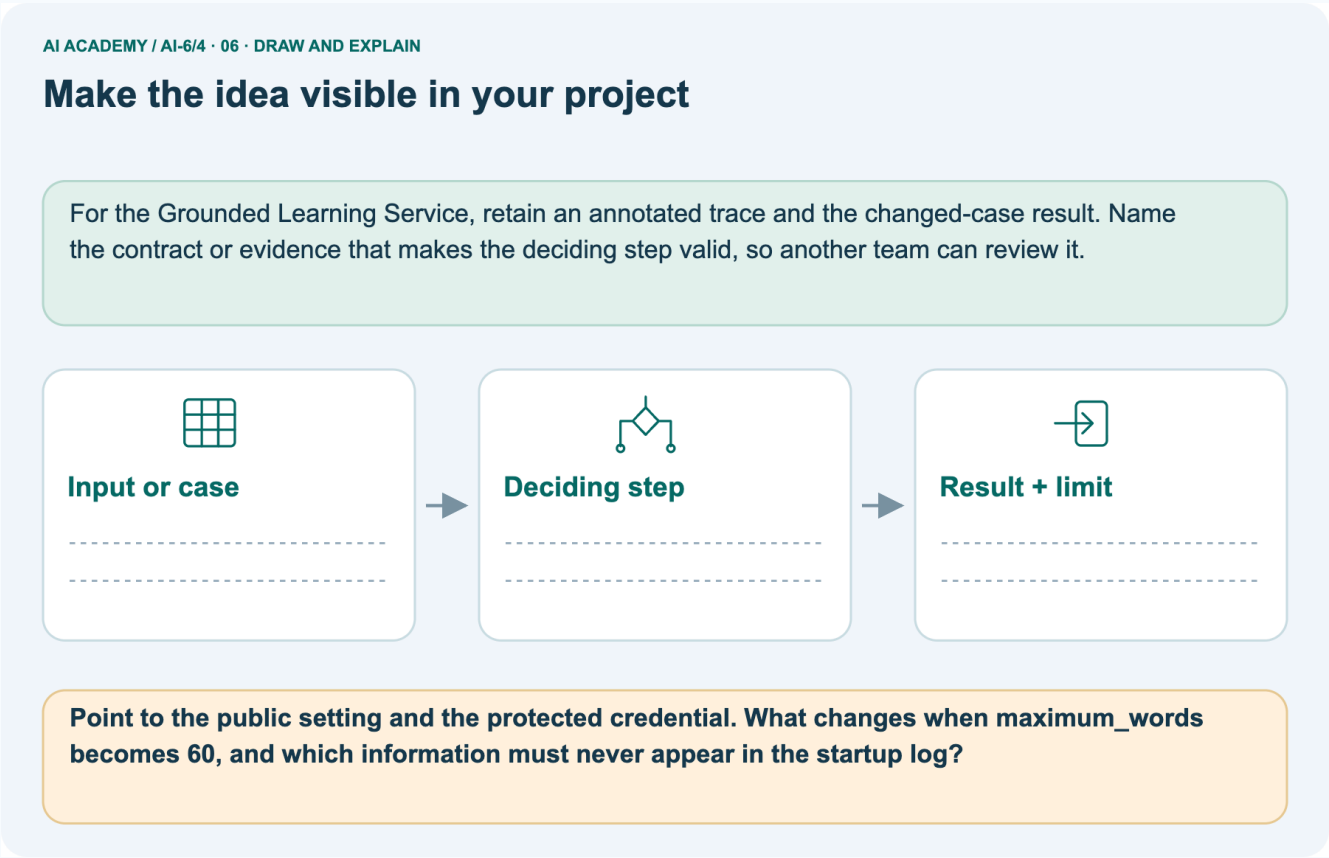
Configuration controls behaviour. A credential grants access and needs different storage and handling. Limit: Protected storage alone cannot prevent leaks through logs, screenshots or excessive runtime permissions.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Protected storage alone cannot prevent leaks through logs, screenshots or excessive runtime permissions.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Point to the public setting and the protected credential.

What changes when maximum_words becomes 60, and which information must never appear in the startup log?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Read the reviewed maximum_words value of 80. The runtime obtains the token through its protected path. Record startup success without printing the token.

Changed case: The configured answer limit becomes 60; the private token stays unchanged. An ordinary length setting does not require replacing the credential.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Configuration and secrets: the case

02. Follow the deciding steps

03. Configuration and secrets — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.